Betweenness Centrality Prediction with Hyper-Connected GNNs

Project Progress Report

Research Log Synthesis

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1 Research Question

Can deep GNNs predict betweenness centrality (BC) inductively, and do HC / mHC / mHC-lite improve depth scaling and out-of-distribution (OOD) ranking quality compared with strong deep-GNN baselines?

Primary evaluation metrics:

- Ranking: Kendall τ, Spearman ρ
- Top-K: Precision@K, NDCG@K

Protocol (inductive): train on small synthetic graphs (100 nodes, ER/BA/SBM), evaluate on larger unseen graphs (500 nodes, same families = ID; WS/RGG families = OOD).

2 Stage 1 — Full 5-Seed Depth Sweep (2026-03-06)

2.1 Setup

Seeds	5 (0–4)
Models	19 variants: GCN, SAGE, GAT, GIN, GCNII, APPNP, JKNet, HC, mHC, mHC-lite
Depths	$L \in \{2, 4, 8, 16, 32\}$
Train graphs	160 (100 nodes, ER/BA/SBM)
Val / Test ID / OOD	30 / 30 / 30 (500 nodes)
Budget	50 epochs, patience 10, LR 10^{-3}, WD 10^{-5}

2.2 Best points from the frozen sweep

Category	Model	L	OOD Kendall
Best overall OOD	`hc_gcnii`	2	0.3824 ± 0.0253
Best mHC point	`mhc_gcn`	2	0.3812 ± 0.0238
Strongest GCN gain	`hc_gcn`	4	0.3809 ± 0.0386
Best deep baseline	`appnp`	32	0.3544 ± 0.0258

2.3 Conclusions

HC / mHC improve OOD consistently; gains concentrate at $L \leq 8$. No ultra-deep scaling claim is supported. Two families retained: **GCN** (clearest mHC signal) and **GCNII** (deep-backbone stress test).

3 Stage 2 — Targeted Hyperparameter Search (2026-03-06)

Optuna proxy search (15 trials, validation-only objective):

- **GCN Trial 3** (obj. 0.8605): LR 2.26×10^{-4}, WD 2.10×10^{-5}, dropout 0.1, $n_{\text{str}} = 2$, $\tau = 0.2$, iters 20.
- **GCNII Trial 11** (obj. 0.8564): LR 1.07×10^{-3}, WD 4.34×10^{-5}, $n_{\text{str}} = 4$, $\tau = 0.1$, GCNII $\alpha = 0.1$, $\theta = 0.5$.

GCNII failure at L16 persists after tuning. GCN + mHC is the strongest candidate.

4 Stage 3 — Multi-Seed Confirmation (2026-03-07)

90 runs per family (3 candidate trials × 3 depths × 5 seeds).

GCN: mHC-GCN consistently gains on OOD Kendall at L4/L8 (strongest: +0.071 at t3/L4). Single negative case: t13/L2 (−0.006).

GCNII: mHC helps at L2/L8 but degrades at L16 (deltas: −0.039, −0.012, −0.047).

Decision: GCN family only from this point.

5 Stage 4 — GCN Ablation Study (2026-03-07)

5.1 Results at L8 (tuned GCN regime, 5 seeds)

Block	Variant	Val Kendall	OOD Kendall
Structure	gcn_ref	0.8332	0.3574
Structure	mhc_gcn_ref	0.8909	0.3888
Structure	**hc_gcn_ref**	**0.9174**	**0.4054**
Structure	dynamic only	0.8882	0.3904
Structure	static only	0.8832	0.3855
Structure	$n = 4$ streams	0.8809	0.3791
Sinkhorn	$\tau = 0.05$	0.8873	0.3923
Sinkhorn	$\tau = 0.10$	0.8932	0.3973
Sinkhorn	$\tau = 0.20$	0.8952	0.3911
Sinkhorn	iters = 10	0.8886	0.3906

5.2 Interpretation

HC-GCN dominates. No mHC variant exceeds HC regardless of τ tuning. $n = 2$ streams is optimal. Dynamic and static components contribute equally. The useful mechanism is flexible multi-stream routing, *not* the doubly-stochastic Sinkhorn constraint.

6 Stage 5 — Final GCN Family Comparison (2026-03-09)

Table 1: Final GCN family — test OOD results (5 seeds)

Variant	L	Kendall	Spearman	P@top5%	NDCG@top5%
gcn	4	0.302 ± 0.045	0.428 ± 0.064	0.302	0.530
gcn	8	0.360 ± 0.021	0.506 ± 0.028	0.315	0.580
hc_gcn	4	0.393 ± 0.031	0.543 ± 0.041	0.341	0.589
hc_gcn	8	$\mathbf{0.406 \pm 0.041}$	$\mathbf{0.558 \pm 0.054}$	0.351	0.588
mhc_gcn	4	0.374 ± 0.043	0.521 ± 0.055	0.329	0.581
mhc_gcn	8	0.396 ± 0.038	0.548 ± 0.049	0.343	**0.590**
mhc_lite_gcn	4	0.376 ± 0.042	0.523 ± 0.055	0.332	0.574
mhc_lite_gcn	8	0.395 ± 0.039	0.544 ± 0.050	0.342	0.581

Key findings:

1. **Val:** HC gains are learned (+10% Kendall, $\sigma \approx 0.002$). Depth effect is neutral for GCN but significant for mHC (L4→L8).
2. **Test ID:** All models hit the same ceiling ($\approx$ 82–83% Kendall). Hyper-connections are irrelevant on known distributions.
3. **Test OOD:** Gains are substantial: HC-GCN > mHC-GCN ≈ mHC-lite-GCN > GCN. On top-K (NDCG@top5%), HC and mHC nearly tie.

7 Stage 6 — Residual Baseline (2026-03-10)

Table 2: Residual baseline — OOD Kendall (5 seeds)

Variant	L	OOD Kendall	Relative gain
gcn	4	0.305 ± 0.046	—
gcn_residual	4	0.352 ± 0.033	+15.2%
hc_gcn	4	0.395 ± 0.037	+29.3%
mhc_gcn	4	0.380 ± 0.049	+24.4%
gcn	8	0.359 ± 0.024	—
gcn_residual	8	0.369 ± 0.035	+2.6%
hc_gcn	8	$\mathbf{0.411 \pm 0.040}$	+14.5%
mhc_gcn	8	0.389 ± 0.046	+8.4%

At L8: residual gives only +2.6%; HC-GCN gives +14.5%. The hyper-connection gain is architectural, not reducible to skip connections.

8 Stage 7 — Zero-Shot Real-Graph Transfer (2026-03-09/10)

8.1 Setup

Models trained only on synthetic graphs evaluated zero-shot on:

- **Cora** (2708 nodes, 5278 edges, $\bar{d} \approx 3.9$)
- **CiteSeer** (3327 nodes, 4552 edges, $\bar{d} \approx 2.7$)
- **PubMed** (19717 nodes, 44324 edges, $\bar{d} \approx 4.5$)

8.2 Results — Kendall τ (mean ± std, 5 seeds)

Variant	Cora	CiteSeer	PubMed
gcn_l8	0.350 ± 0.053	0.067 ± 0.118	0.539 ± 0.025
gcn_residual_l8	0.355 ± 0.062	0.069 ± 0.080	0.605 ± 0.042
hc_gcn_l8	0.366 ± 0.034	0.056 ± 0.029	0.595 ± 0.028
mhc_gcn_l8	$\mathbf{0.405 \pm 0.036}$	$\mathbf{0.151 \pm 0.083}$	0.598 ± 0.030
mhc_lite_gcn_l8	0.380 ± 0.037	0.125 ± 0.085	$\mathbf{0.611 \pm 0.030}$

8.3 Per-graph interpretation

Cora ($\bar{d} \approx 3.9$). All hyper-connected variants improve. BC is not severely skewed. mHC-GCN leads on all metrics (+15.5% Kendall, +43.8% P@top5%). HC also gains (+4.5%) without regression.

CiteSeer ($\bar{d} \approx 2.7$ — extreme sparsity). Most informative graph. BC concentrates on a small number of bridge nodes. HC-GCN *regresses* -16% in Kendall vs. plain GCN, while achieving best NDCG@10 (+34%): HC learns to route information toward bridges (correct top-K) but destroys global ranking of other nodes. The Sinkhorn constraint of mHC prevents this collapse: +127% Kendall gain.

PubMed (19717 nodes). Gains more moderate. GCN+residual is very competitive (+12.2%). mHC-lite leads (+13.3%): full Sinkhorn overhead not worthwhile at this scale; lighter variant offers better trade-off. Precision@10 saturates at 0.5 for all models (non-discriminative).

8.4 Routing collapse on sparse BC

The HC regression on CiteSeer reveals a regime-dependent property:

HC (unconstrained)	**mHC (Sinkhorn)**
Best on synthetic BC (moderate skew)	Best on real citation graphs
High routing freedom	Proportional stream balancing
Collapses on sparse/skewed BC	No regression on any tested graph

9 Medium-Term Conclusions

9.1 What the project has established

1. **Hyper-connections improve inductive BC ranking.** Every HC-family variant outperforms GCN on OOD synthetic and on real-graph zero-shot transfer. The gain is architectural, not reducible to residual connections.
2. **The gain is OOD-specific.** On in-distribution test graphs all models hit the same ceiling. Hyper-connections transfer knowledge; they do not add raw capacity on known distributions.
3. **HC-GCN is the best variant on the synthetic protocol.**

$$\text{HC-GCN } (0.406) > \text{mHC-GCN } (0.396) > \text{mHC-lite-GCN } (0.395) > \text{GCN } (0.360)$$

4. **mHC-GCN is the most robust variant in zero-shot transfer.** It never regresses below the GCN baseline across any benchmark. HC-GCN collapses on very sparse, skewed-BC graphs (CiteSeer). The Sinkhorn constraint acts as routing regularization.
5. **Deep scaling is not validated.** GCNII + mHC fails at L16. HC-family advantages concentrate at $L \leq 8$ for GCN backbones.

9.2 What cannot be claimed yet

- mHC does not universally outperform HC.
- Deep scaling ($L > 8$) is not validated for any variant.
- The HC routing collapse is a single observation (CiteSeer); additional sparse graphs are needed to generalize it.
- No parameter-count-matched comparison exists; gains could partially reflect extra capacity.

9.3 Recommended next steps

1. **BC spectrum analysis:** correlate BC distribution skewness (Gini, power-law exponent) of each graph with the HC vs. mHC relative performance, to test the routing-collapse hypothesis.
2. **Internal matrix analysis:** inspect inter-stream transport matrices of HC-GCN-L8 vs. mHC-GCN-L8 on CiteSeer to characterize the collapse geometrically.
3. **Fair capacity comparison:** add a parameter-count table and capacity-matched baselines.
4. **More sparse real graphs:** test on additional sparse social or collaboration networks to generalize the CiteSeer finding.